

Countability

• Have some infinite sets: $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$

- doesn't mean same cardinality
- different "classes" of infinity

• From functions (see lecture 9):

- A and B have same cardinality ($|A| = |B|$) iff we can construct bijection from A to B
- Bijection means one-to-one and onto
- One-to-one means outputs not repeated
- Onto means outputs cover entire co-domain

ex. Provide bijection $f: \mathbb{N} \rightarrow \mathbb{Z}$ that shows $|\mathbb{N}| = |\mathbb{Z}|$

$$\text{Define } f: \mathbb{N} \rightarrow \mathbb{Z} \text{ s.t. } f(n) = \begin{cases} \frac{n}{2} & \text{when } n \text{ even} \rightarrow \{0, 1, 2, 3, \dots\} \text{ non-negative } \mathbb{Z} \\ -\frac{(n+1)}{2} & \text{when } n \text{ odd} \rightarrow \{-1, -2, -3, \dots\} \text{ negative } \mathbb{Z} \end{cases}$$

*don't need full proof, just need to provide a bijection

f covers all $\mathbb{Z}$, no outputs repeated $\therefore f$ bijection

Since we found bijection $f: \mathbb{N} \rightarrow \mathbb{Z}$, then $|\mathbb{N}| = |\mathbb{Z}|$

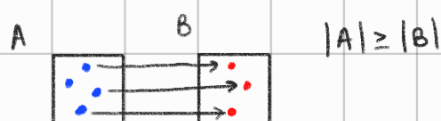
• **Countably Infinite:** set A is countably infinite iff there is a bijection from $\mathbb{N} \rightarrow A$

$$\text{eg. } \begin{array}{ccccccc} \mathbb{N}: & 0 & 1 & 2 & 3 & \dots \\ & \downarrow & \downarrow & \downarrow & \downarrow & \\ \mathbb{Z}: & 0 & -1 & 1 & -2 & \dots \end{array}$$

Can count integers in some ordered fashion indexed by $\mathbb{N}$

Too hard to produce a bijection?

• Cantor Schröder-Bernstein Theorem: $|A| \leq |B|$ iff $\exists$ a one-to-one function from A to B



- So if we can find $f: A \rightarrow B$ one-to-one and $g: B \rightarrow A$ one-to-one, $|A| \leq |B|$ and $|A| \geq |B|$ thus $|A| = |B|$ (two way bounding)

ex. Provide two one-to-one functions to show that the non-negative rationals are countably infinite

$$\left. \begin{array}{l} f: \mathbb{N} \rightarrow \mathbb{Q}^{\geq 0}, \quad f(n) = n \\ g: \mathbb{Q}^{\geq 0} \rightarrow \mathbb{N}, \quad g\left(\frac{x}{y}\right) = 2^x 3^y \end{array} \right\} \rightarrow |\mathbb{N}| = |\mathbb{Q}^{\geq 0}|$$

$\downarrow$
rational number
n in lowest terms

19.2 A Curious Bijection

Recall that two sets have the same size (cardinality) if there is a bijection between them. You might imagine that $\mathbb{N}^2$ has more elements than $\mathbb{N}$. However, in this problem we will investigate a function $f : \mathbb{N}^2 \rightarrow \mathbb{N}$ that turns out to be a bijection and, therefore, shows these two sets are, in fact, the same size.

Before defining f , let us agree to denote by $s(n)$ the sum of the first n natural numbers. In other words, s is the following function:

$$s : \mathbb{N} \rightarrow \mathbb{N} \text{ by } s(n) = \sum_{i=0}^n i$$

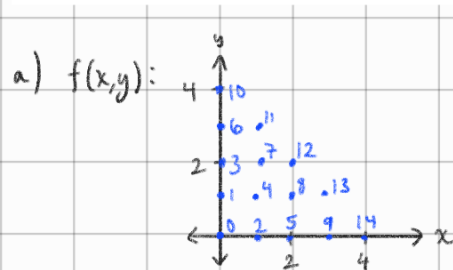
Now we define f in terms of s as follows:

$$f : \mathbb{N}^2 \rightarrow \mathbb{N} \text{ by } f(x, y) = s(x + y) + x$$

The following questions will explore some properties of this seemingly complicated function:

- Draw a picture of what the function does for pairs (x, y) such that $x + y \leq 4$. Specifically, draw the usual xy -coordinate plane, and, at the (x, y) -position, write the output value $f(x, y)$. For example, the location $(1, 2)$, should contain the value 7.
- For some fixed natural number k , consider all pairs (x, y) with $x + y = k$. What range of output values does f produce for this set of input pairs?
- What is the preimage of the output value 17?
- Suppose that we have two pairs (x, y) and (p, q) which are not the same and for which $x + y \neq p + q$. Explain why $f(x, y) \neq f(p, q)$.
- Suppose that we have pairs (x, y) and (p, q) which are not the same but for which $x + y = p + q$. Explain why $f(x, y) \neq f(p, q)$.

The answers to (d) and (e) could easily be combined into a formal proof that f is one-to-one. Similarly, the pattern observed in (a), (b), and (c) leads to a proof that f is onto.



b) max when $y=0, x=k$, min when $x=0, y=k$ so
range = $[k!, k! + k]$

c) $x=2, y=3$

d) f is one-to-one, so $f(x, y) \neq f(p, q)$

e) same reasoning